MODEL REFERENCE

A one-dimensional radiative-convective atmospheric column

17,200 lines of C# that march a 50 km column of air to radiative–convective equilibrium, with longwave opacity built from real HITRAN line data rather than a tuned grey constant — and with the far wings of CO_2's 15 µm band corrected for the sub-Lorentzian shape that sets its forcing.

What it computes

Given a solar input, a planetary albedo and an absorber loading, it solves for the temperature at every level of an atmospheric column and at the surface beneath it — the state in which the column radiates to space exactly what it absorbs, and no level is left with a net heating tendency.

Two balances have to close at once, and they are not independent. At the top of the atmosphere the outgoing longwave must equal the absorbed sunlight. At the surface, the absorbed sunlight plus the back radiation must equal what the ground emits plus what convection and evaporation carry away. A column can satisfy either alone; equilibrium is where both hold together.

SURFACE TEMPERATURE	OUTGOING LONGWAVE	GREENHOUSE EFFECT	CONVECTING LAYER
286.797 K	238.175 W m⁻²	137.8 W m⁻²	3.44 km
13.65 °C, default configuration	equal to absorbed solar at equilibrium	surface emission minus outgoing longwave	above it, radiation alone

Everything below is how those numbers are arrived at, and — equally — which of them are predictions and which are calibrations. That distinction is kept explicit throughout, because a model of this kind can be made to produce almost any answer if the difference is allowed to blur.

The vertical grid

The column is divided into segments of equal geometric thickness from the surface to 50 km. Each carries a pressure at its lower and upper interface, a mid-layer pressure, and an air mass per unit horizontal area obtained from hydrostatic balance:

$$dp = -\rho\ g\ dz$$

so the mass supporting a pressure drop is $\Delta p/g$, independent of the density profile

> Pressures come from the US Standard Atmosphere. That standard is defined on *geopotential* altitude, which measures height in units of work done against a constant reference gravity, not in metres. The two differ by
>
> > $$H = r_0 z / (r_0 + z)$$
> >
> > about 140 m at 50 km on Earth
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> and the model converts between them rather than treating the standard's tabulated heights as geometric. Ignoring the distinction would put the top of the column in the wrong place by more than the thickness of a segment.
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> Every flux in the model is **power per unit area of planet surface**, including inside the atmosphere. That convention is what makes the energy budgets close arithmetically, and it is what allows spherical geometry to be added later as a single scaling rather than a rewrite.

Longwave emission: the Koenigsberger equation

A volume of gas at temperature T emits

$$dq = 4\ \varepsilon'\ \sigma\ T^4\ dV$$

ε' is a volumetric emission coefficient, m^{-1}

The factor of four is not decorative: it is the ratio of the surface area of a sphere to the area of its shadow, and it is what converts a directional emission into an isotropic one. Because every quantity here is per unit horizontal area, dV reduces to a thickness dz, and a segment's total emission is $4\ \varepsilon'\ \sigma\ T^4\ dz$.

This is the optically thin limit. The solver does not use it directly — it uses the exponential form below, which reduces to exactly this as $dz \to 0$, and the model checks that it does.

Why the diffusivity factor is exactly two

Radiation crosses a slab in every direction, not just vertically. A ray at zenith angle θ travels a path $1/\cos\theta$ longer than the vertical one, so a flux calculation has to integrate over the hemisphere. That integral gives the exponential integral E_3, and the transmittance of a slab of vertical optical depth τ is $2E_3(\tau)$.

Two-stream models replace that integral with a single effective path, D times the vertical one. The usual choices are $D = 1.66$ (Elsasser) or 2. This model takes $D = 2$, and the reason is that it is the only value consistent with the emission law above:

$$2E_3(\tau) \to 1 - 2\tau \quad \text{as } \tau \to 0$$

so absorptivity → 2τ, matching the $4\varepsilon'\sigma T^4$ emission exactly

Choosing $D = 1.66$ would make the column absorb 17 % less than it emits in the thin limit, which violates Kirchhoff's law at the level of the model's own construction. The better far-field accuracy of 1.66 is real, but it is bought by breaking an identity the rest of the model depends on. D is a parameter; the default is the consistent one.

The two-stream recurrence

Upward and downward fluxes are marched across segments. For a segment of hemispheric optical thickness $\tau = D\,\varepsilon'\,dz$, with transmittance $t = e^{-\tau}$:

$$F_{i+1}^{\uparrow} = t\,F_i^{\uparrow} + (1 - t)\,\sigma T_i^4$$

and symmetrically downward, from the top

This is exact, not a discretisation. For a segment at constant temperature the Schwarzschild equation integrates to precisely this expression, which has a consequence worth stating: subdividing an isothermal slab leaves its emission unchanged. Grid refinement in this model therefore tests the *temperature structure*, not the radiative transfer, and any convergence error that shows up is about where the temperature is sampled rather than about how radiation is propagated.

The boundary conditions are that nothing enters the top of the column, and that the surface emits $\varepsilon_s\sigma T_s^4$ and reflects the fraction $(1 - \varepsilon_s)$ of the back radiation reaching it.

From a grey slab to real spectroscopy

A grey model gives every wavelength the same opacity. Real gases do not: CO_2 is nearly opaque at 15 μm and transparent at 10, and that difference is the whole reason the response to more CO_2 is logarithmic rather than linear.

The model can be run either way. In its spectral mode it takes HITRAN line lists for six absorbers — water vapour in two bands, CO_2's 15 μm band, ozone at 9.6 μm, methane at 7.7 μm and nitrous oxide at 7.8 μm — and derives bands from the line data itself:

1. **Resolve the spectrum.** Each line is given a Lorentz profile with half-width $\gamma = \gamma_{ref}(p/p_{ref})(296/T)^n$, where *n* is HITRAN's own temperature exponent. Collisions are more frequent at higher pressure and faster at higher temperature, and the two pull the width in opposite directions — a stratospheric layer at a tenth of an atmosphere and 220 K comes out about 12 % broader than pressure alone would give.
2. **Split into bands.** Band edges are placed so each carries a comparable share of the Planck function at a representative temperature, rather than being equally spaced in wavenumber.
3. **Measure a k-distribution per band.** The resolved absorption coefficients inside a band are sorted and reduced to a small number of quadrature points — the correlated-k method. Sorting is what makes this work: the reordered spectrum is smooth, so a handful of points integrates it accurately where thousands would be needed in wavenumber order.

All six gases are derived on one shared wavenumber grid. Deriving them separately would produce overlapping band sets — N_2O's 7.8 μm band sits inside methane's, and water vapour is everywhere — and overlapping bands each claim their own share of every emitter's Planck function, so the spectrum would sum to more than one.

The far wings, and the correction that changed the answer

The Lorentz profile comes from the impact approximation: collisions treated as instantaneous compared with the time between them. That holds near line centre and fails far from it. At a detuning of tens of cm^{-1} the radiation is probing the collision itself, over times short enough that its finite duration matters, and real CO_2 wings fall off **faster** than Lorentzian.

That is not a detail here. This model's CO_2 forcing comes almost entirely from the far wings — the band core is saturated, so only the wings can still respond to more gas — which is precisely *why* the response is logarithmic. The far-wing shape is the thing that sets the forcing coefficient.

The evidence arrived before the explanation. A convergence study over the wing cutoff gave a forcing coefficient of 5.15, 6.63, 6.88 and 6.98 W m^{-2} per ln as the cutoff opened from 100 to 800 cm^{-1} — the answer was being set by how far out the wings were integrated, and it converged on a value about 1.30 times the accepted 5.35.

The correction is the three-segment exponential of Perrin & Hartmann (1989), multiplying the Lorentz profile:

$$\chi(s) = \exp[-B_1(s-s_1) - B_2(s-s_2)^+ - B_3(s-s_3)^+]$$

$s = |\nu - \nu_0|$, breakpoints at 3, 30 and 120 cm^{-1}, continuous at each

It is applied to **CO_2 alone**. A χ factor is fitted to one band of one gas; giving CO_2's to water vapour would be inventing spectroscopy rather than correcting it. It is also applied *outside* the k-distribution, because droplet-free line absorption and the χ correction are different functions of wavenumber and folding them together would misplace the correction.

	ABSORBER SCALE	BASE T_S	A	VS ACCEPTED
Pure Lorentz wings	14.578	286.796 K	6.988	1.31×
χ factor, same loading	14.578	285.109 K	5.073	0.95×
χ factor, recalibrated	15.869	286.796 K	5.091	0.95×

Measured in two steps, so the shape change and the loading change are not confounded

Essentially the entire over-forcing was that one piece of missing spectroscopy. Two things are worth reading off the table. Suppressing the wings removes absorption, so the base state cooled 1.7 K and the gas loading had to come back up — but **recalibrating barely moved the coefficient**, 5.073 to 5.091, which is the reassurance that it is a property of the spectroscopy and not of how much gas was put in.

A caveat that travels with these numbers. The functional form is Perrin & Hartmann's; the coefficients are representative values for the CO_2 ν_2 band near 296 K and have *not* been checked against the original paper. This is a correctly shaped correction rather than a faithful reproduction of a published fit. Nothing was tuned to reach 5.35 — the coefficient is reported as it came out, which is the only way the forcing stays a prediction.

Why the response is logarithmic

Doubling the CO_2 doubles the absorber, but does not double the forcing. The reason is spectral saturation, and it can be seen in one line of algebra.

At the centre of the 15 μm band the atmosphere is already opaque: adding gas there changes nothing, because no more radiation can be stopped. The only place extra CO_2 can act is in the wings, where the band is still partly transparent. So the forcing is set by how fast the *opaque width* of the band grows.

If absorption falls off exponentially with distance from band centre, $k(\nu) \approx k_0 e^{-|\nu-\nu_0|/a}$, the band is opaque wherever $k\,u > 1$, which happens out to

$$W = 2a \ln(k_0\, u)$$

so $F \propto \ln u$ — the logarithm, in one step

The shape of the wing is doing all the work, and a different wing gives a different law. A Lorentzian wing, falling as $1/(\nu-\nu_0)^2$, would give $W \propto \sqrt{u}$ and hence a square-root response. A Gaussian wing would give $\sqrt{(\ln u)}$. Only something close to exponential gives a logarithm.

Which leads to a result worth stating plainly: correcting the wings made the model less logarithmic, not more. Drift of the fitted coefficient across the sweep went from 2–5 % to 8–10 %. That is not a regression — it follows directly from the algebra above. The pure Lorentzian far wing, integrated to a wide cutoff, happens to produce a very clean logarithm; suppressing it with a χ factor breaks that idealisation. The model was more logarithmic when it was more wrong.

The grey configuration makes the contrast concrete. Given one opacity for the whole spectrum, its forcing ratio against the accepted law grows from 1.17 to 2.14 across the same concentration span, because its absorber is simply diluted rather than spectrally resolved. There is no band core to saturate, so there is no logarithm to recover.

Water vapour, and its feedback

Water vapour is the largest greenhouse gas in the column and the only absorber whose amount follows the temperature. Its loading is scaled by Clausius–Clapeyron integrated at constant latent heat,

$$e_{sat}(T) = e_0 \exp[(L/R_v)(1/T_0 - 1/T)]$$

evaluated at the near-surface air temperature

and distributed with its own scale height rather than following air density. Because it is re-evaluated before every radiation solve, a warming column becomes more opaque as it warms, which is the water-vapour feedback.

> The feedback can be switched off, and the comparison is instructive — but only if it is set up carefully. Freezing the vapour at its nominal loading gives a *different base state*, so the two runs would start 2 K apart and the comparison would measure that offset. The no-feedback configuration is therefore frozen at the feedback run's own base air temperature, so both begin at exactly 286.796 K.
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> Set up that way, the two produce **identical forcings** — to four decimal places — and different warmings. That identity is not a coincidence: instantaneous forcing is defined at held temperatures, so a feedback cannot act on it. It is the cleanest demonstration in the model that a feedback changes the response and not the forcing.
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> > The feedback is a demonstration of mechanism, not a quantitative estimate. It scales the whole vapour column by one Clausius–Clapeyron factor taken at the surface, with a fixed exponential shape, so it cannot represent upper-tropospheric humidity — which is where the outgoing longwave is actually set. And the prescribed lapse rate means nothing can offset it, where a real lapse-rate feedback would cancel perhaps a third of it.

Clouds

Clouds do two opposite things, and the model does both. They are off by default.

In the shortwave they reflect. The albedo splits into a clear part and a cloudy part, mixed by cloud fraction:

$$A = (1-f)\,A_{clear} + f\,A_{cloud}$$

0.155 and 0.361 over 67 % of the sky returns Earth's 0.293

That split is what stops the clouds being counted twice. Earth's 0.30 planetary albedo *already contains* them; a reflective deck added on top of it would reflect the same sunlight again.

In the longwave they trap. The deck is specified by the emissivity it should have as a whole, inverted to an optical thickness $\tau = -\ln(1-\varepsilon)$ and spread over the segments by their *overlap* with the cloud rather than their full thickness — so a 1.0–4.5 km deck keeps the same emissivity however the column is divided up. Its opacity is grey, and is added outside the k-distribution: droplets absorb across a band where gas absorbs in lines. Folding it inside would have made the cloud thin wherever the gas is transparent, which is the exact opposite of what a cloud does, since its longwave work happens mostly in the window.

Two skies, one atmosphere. The longwave is solved twice, with and without the deck, and the fluxes mixed by cloud fraction — the independent column approximation. The mixing is exact, since fluxes add. What is approximate is the premise: that the cloudy and clear parts of the sky exchange no radiation sideways. Both solves see the same temperatures, because there is one atmosphere and one surface under both skies.

	MODEL	CERES
Shortwave — reflected away	−46.96	−47.1
Longwave — trapped	+26.55	+26.2
Net	**−20.41**	**−20.9**

Cloud radiative effect, W m^{-2}, against the CERES satellite record

Only the longwave figure is a result; the shortwave one is arithmetic on the two albedos, and those were chosen to reproduce Earth's. Both are measured as CERES measures them — this planet against the same planet with the cloud removed and nothing else touched.

The calibration is separate, and finding out why was the interesting part. Switching clouds on over the shipped configuration warms the surface by about 13 K. Nothing is broken: that configuration's absorbers were scaled to reach an Earth-like temperature at an 0.30 albedo with *no cloud* — a planet carrying the clouds' reflection while having none of their greenhouse. The gas had been standing in for the cloud greenhouse all along, so adding a real deck supplies it twice. The cloudy configuration therefore halves the gas loading and hands the difference to the deck, arriving at the same surface by a different route.

A second result fell out of that. Running the four corners — each effect on and off — the two nearly cancel in temperature while emphatically not cancelling in flux: reflection cools by 12.45 K, trapping warms by 11.75 K, and the net is **−0.99 K** against a net flux effect of −20.41 W m^{-2}. A single sensitivity applied to that flux would have predicted twelve. The components have different efficacies here — 0.27 against 0.44 K per W m^{-2} — which is a property of this model's fixed lapse rate and prescribed deck rather than a claim about Earth. It does illustrate why cloud feedback is the largest remaining uncertainty in real climate sensitivity: a net number that small is a difference between two much larger ones.

Convection and the surface

Radiative equilibrium alone gives a surface far hotter than observed and a lapse rate steeper than the atmosphere can sustain. Three non-radiative terms fix that.

A critical-lapse-rate adjustment. Wherever the lapse rate exceeds 6.5 K km^{-1} the affected block is mixed back onto exactly that rate, conserving $\Sigma\ C_i T_i$. This is not a diffusion — it is an instantaneous restoration of the profile a convecting atmosphere would hold, and it is applied after every timestep.

A surface sensible heat flux, $Q = h_c(T_s - T_{air})$, with $h_c = 5.8 + 4.1v$. The air temperature is extrapolated to z = 0 from the two lowest segments rather than taken from the lowest one, which matters more than it sounds: using the lowest segment evaluates the air at *dz*/2 and makes the whole model first order in *dz*. The radiative recurrence is already exact for constant-temperature segments, so this extrapolation is what restores second-order convergence.

An optional latent heat flux — evaporation — through the bulk aerodynamic form, off by default. Its moisture-availability parameter β scales the *whole flux*, not the surface humidity; scaling the humidity instead makes β < 1 drive perpetual dew deposition and *warm* the surface, which is the opposite of what suppressing evaporation should do.

The surface is deliberately excluded from the lapse-rate adjustment. It exchanges heat with the air through the flux above and through radiation, and folding it into the convective mixing would let it equilibrate with the air instantly, removing the temperature difference that drives the flux in the first place.

Geometry, optionally

Three corrections take the column from a plane-parallel idealisation toward a real planet. Each is switchable, and their sizes are worth knowing because two of them nearly cancel.

Sphericity. A shell at radius *r* has $(r/r_0)^2$ times the area of the surface beneath it. Writing $G = (r/r_0)^2F$ — the power crossing radius *r*, divided by the area beneath it — removes the geometric term from the spherical two-stream equations entirely, leaving the plane-parallel equation with its source scaled by $(r/r_0)^2$. So sphericity enters the solver in exactly one line, the recurrence stays exact, and every budget still closes in W per m^2 of surface. Effect: −0.016 K.

Gravity falling with height. $g(z) = g_0(r_0/(r_0+z))^2$, about 1.5 % weaker at 50 km, which changes the mass supporting each pressure drop. Effect: +0.012 K.

The top-of-atmosphere solar cross-section. The planet intercepts sunlight on a disc of radius $r_0 + z_{top}$, not r_0, and rays near the limb pass through a long slant path without ever reaching the ground. Effect: +0.309 K — an order of magnitude larger than the other two, and the only one that materially matters.

Solving it

The column is marched to equilibrium with an adaptive explicit forward-Euler step on each segment's heat capacity. The step is limited by two constraints: no level may move more than a fixed amount in one step, and the step must stay inside the explicit stability limit set by the local radiative relaxation time C/λ, where λ is the derivative of net loss with respect to temperature.

That second constraint is what makes the problem stiff. The thin upper segments have small heat capacities and short relaxation times, so they set a step much shorter than the lower column needs, and a single march takes some 44,000 steps. It is the dominant cost of the model.

Convergence is measured across the *complete* step, adjustment included. Inside a convecting block an individual segment's radiative tendency does not vanish at equilibrium — it is balanced by the convective adjustment — so the tendency alone is not a convergence measure.

A concentration sweep runs twenty of these marches. Every concentration above the reference is independent of the others — each builds its own column and measures its forcing against the one shared reference state — so they run concurrently, and the sweep the charts need went from 217 seconds to 59 on a 32-core machine. That is a scheduling change and not a numerical one, and the full-precision output confirms it: every value identical to the digit.

What is calibrated, and what is predicted

This is the section that decides how much weight anything else can carry.

Calibrated. The absorber amounts are scaled by one common factor chosen to put the base state at an Earth-like surface temperature. The continuum that closes the atmospheric window is one tuned number rather than fitted MT_CKD coefficients. The cloud albedos and fraction are chosen to reproduce Earth's clear-sky and all-sky albedos. The cloud deck's height was chosen by requiring the longwave cloud effect to match observation. In the grey configurations, the CO_2 share of the absorber is tuned against a known forcing.

Predicted. Which bands exist, how opaque each is relative to the others, and how absorption is distributed within each — all of that comes from the line data. So does the *shape* of the CO_2 response: the logarithm is not imposed anywhere, and the absorber amount is exactly linear in concentration, so the curvature is produced entirely by band structure. The forcing coefficient that comes out of it is a measurement, reported as it falls.

The distinction matters most for the headline comparison. Because the CO_2 forcing is measured in W m^{-2} and compared against a law also expressed in W m^{-2}, **nothing is converted and no sensitivity is borrowed**. Turning the accepted law into a temperature would need the model's own sensitivity, which would make the reference partly a restatement of the thing it is meant to test — which is why the temperature views carry no reference curve at all.

Results

	CLEAR SKY	67 % CLOUD
Absorber scale	15.869	5.142
Base surface temperature	286.796 K	286.796 K
Forcing at 580 ppm	3.719 W m⁻²	3.336 W m⁻²
Forcing at 1000 ppm	6.238 W m⁻²	5.854 W m⁻²
Fitted coefficient A	4.839	4.676
Against accepted 5.35	0.904×	0.874×
Drift across the sweep	10.1 %	1.1 %
Warming to 1000 ppm	+4.35 K	+3.45 K
— with vapour held fixed	+2.33 K	+2.54 K
Sensitivity	0.708 K/W m⁻²	0.592 K/W m⁻²
Convective top, 285 → 1000 ppm	4.17 → 5.83 km	5.83 → 5.83 km
Emission-temperature level	4.53 → 5.17 km	4.24 → 4.76 km

Spectral configuration, 16 bands × 16 g-points, 400 cm^{-1} wing cutoff, χ-corrected

The vertical numbers are the mechanism made visible. Adding CO_2 lifts the height at which the column reaches the planet's emission temperature, and the surface is warmer than that level by the lapse rate times its height. That is the greenhouse argument, measured rather than asserted.

A cloud deck masks about 6 % of the CO_2 forcing — it already blocks part of the upward flux the extra gas would have intercepted. The clouded response is also far closer to a pure logarithm, but holding the base state fixed required dropping the gas loading threefold, so cloud and loading moved together and that one is suggestive rather than established.

How it is checked

381 tests, and the ones worth naming are those that check the model against something outside itself.

- **Against a resolved spectrum.** The correlated-k reduction is checked against direct line-by-line integration of the same spectrum, so the quadrature is verified rather than assumed.
- **Against analytic limits.** The exponential recurrence is checked to reduce to the Koenigsberger form as $dz \to 0$; area under a broadened line is conserved; a spherical shell's emission is checked against an

independent RK4 integration of the spherical two-stream equations.

- **Against observation.** The cloud radiative effect against CERES; the Bowen ratio against Earth's global mean when evaporation is enabled.
- **Convergence rather than agreement.** Grid, band count, g-points and wing cutoff are each refined and the answer's movement recorded. This is what identified the far wings as the dominant error, and what exposed an earlier configuration that had got the right coefficient by two errors cancelling.
- **Invariants that catch structural mistakes.** Forcing is zero at the reference concentration by construction, at every cloud fraction — an assertion that cost nothing and later caught a real bug, where an all-sky baseline was being differenced against a fully cloudy perturbed state.

Continuous integration builds on Windows and on Linux, the latter restoring with no package source at all, so a dependency that quietly crept in would fail the build rather than the user.

What it is not

One column, globally averaged, with no horizontal transport, no diurnal or seasonal cycle, and no dynamics. It cannot produce a circulation, a latitude gradient, or anything that depends on one.

- **The absorber amounts are illustrative.** They are scaled to reach an Earth-like surface, not taken from observed concentrations, so the CO_2 column is not Earth's. The band is more saturated than the real one, and what that does to the forcing coefficient has not been isolated — it is a genuine unknown inside the quoted 0.90×.
- **Line strengths are not temperature-scaled.** Half-widths are, through HITRAN's n_{air}, but intensities are used at their 296 K values because scaling them needs total internal partition sums. Line mixing, Voigt profiles, self broadening and pressure shift are all absent.
- **The instantaneous forcing is not the adjusted forcing.** The accepted $5.35 \ln(C/C_0)$ is a stratosphere-adjusted value at the tropopause; this is an instantaneous one at the top of the atmosphere. They are different quantities, and the comparison should be read with that in mind.
- **One grey cloud deck** at one height stands in for everything from fog to cirrus, and is prescribed rather than predicted — so it cannot act as a feedback, which is the single largest uncertainty in real sensitivity estimates.
- **The shortwave is a deposition profile, not a transfer calculation.** There is no scattering anywhere in the model; cloud and surface reflection are albedos.
- **No feedbacks except water vapour.** No ice–albedo, and no lapse-rate feedback beyond what the fixed critical lapse rate already imposes.

What it is for is showing that the qualitative behaviour of a radiative–convective atmosphere — a convecting troposphere under a radiative stratosphere, a greenhouse effect that scales with the height of the emission level, and a logarithmic response to CO_2 — follows from a small number of physical statements applied carefully, and can be traced from HITRAN line data to a surface temperature without any step being taken on trust.

17,198 lines of C# in 63 files — 5,788 of physics in `ClimateColumn.Core`, 39 % of it comment, and 8,521 lines of tests. 381 tests pass. Longwave opacity is derived from HITRAN line data, which is not redistributed with the source.

`github.com/MarkAndersUK/ClimateColumn`